

Supplementary Material for “Rotating Away the Location Degeneracy: An Exact Reparameterization for Hamiltonian Monte Carlo in Hierarchical and Joint Models”

Section, table, figure and equation numbers without the prefix S refer to the main article.

S1 A Gaussian check of the blocked-sampler comparison

Section 5 states two things about α : that the lag-1 autocorrelation of an exact two-block sampler equals R^2 for a Gaussian posterior (Proposition 5(iii)), and, heuristically, that HMC pays for the same correlation at most in integration steps. We checked both in a Gaussian model, comparing a simulated two-block Gibbs sampler with a stylized model of HMC (exact Hamiltonian dynamics, random integration time). The two-block sampler’s lag-1 autocorrelation matched R^2 to within 0.004, and its integrated autocorrelation time went 3.0, 9.0, 27.6, 39.0 and 199 as R^2 went 0.5, 0.8, 0.93, 0.95 and 0.99. The HMC model’s stayed between 1.6 and 2.1, at a cost of 4.8 to 46.8 gradient evaluations per independent draw. With an unrelated narrow ridge already present (such as, in a joint model, the spline coefficients or variance components might provide), the step count was 22.2 at every R^2 from 0 to 0.99.

S2 Random intercept only: runs to convergence

The same simulator as in Section 6 with $\sigma_1 = 0$, fitted with a random intercept only, gave seven designs: about 9, 17 and 35 visits, $N \in \{100, 300\}$, and one noisier design with $\sigma_e = 0.8$. The intercept gained 6.5–282 times and the subject-constant coefficient of age 6.6–207 times; the time slope, which has no absorbable direction when there is no random slope, moved only through the step size (0.79–3.4 times). At this run length, however, 17 of the 21 unrotated fits had some parameter with $\hat{R}$ above 1.05, the intercept’s reaching 1.27, while every rotated fit had $\hat{R} \leq 1.004$, so these ratios compare a converged

fit with unconverged ones. Posterior means agreed to a median of 0.025 posterior standard deviations (maximum 0.175).

To obtain a comparison between converged fits, we reran three designs ($N = 300$ with about 9, 17 and 35 visits) and extended the unrotated fit to 2, 4, 8, 16 and 32 times the standard length, with two seeds taken to the longest runs. A convergence criterion was fixed in advance: split $\hat{R} \leq 1.01$ on every monitored parameter and $\text{ESS} \geq 400$ for the intercept, age, the time slope and α . Every rotated fit met it at the standard length. The unrotated fit met it first at 16 times the standard length with 9 visits, at 16 and 32 times with 17 visits, and not by 32 times with 35 visits, where its intercept had an ESS of 304–377 from 64,000 draws with $\hat{R} = 1.002$. Its ESS per draw for the intercept did not change with run length: 0.009–0.023, 0.007–0.020 and 0.002–0.008 in the three designs over runs from 1 to 32 times the standard length, against 2.1–2.5, 0.9–1.9 and 1.1–2.4 for the rotated fit. The per-draw advantage, of order 100 times, is therefore a property of the stationary chains, not of short runs. In wall time, the unrotated fit needed 3.5 and 6.5–6.6 times as long as the rotated fit to meet the criterion in the first two designs, and more than 10–14 times in the third. These ratios are much smaller than the per-draw ones because the rotated fit’s time at the standard length is mostly warm-up and compilation, and it exceeded the ESS target by a factor of 5 to 12. Between the first fits of each arm to meet the criterion, posterior means agreed to a median of 0.02 posterior standard deviations (maximum 0.11) and standard deviations to within 6%.

S3 Stress grid

The stress grid of Section 6.4 crossed about 35 visits with $N \in \{50, 100\}$ and $\rho \in \{0.3, 0.95\}$, and added $\rho = 0.95$ at 9 visits ($N = 100$) and at 17 visits ($N = 300$). The criterion fixed before the run was that the rotation must not fall below the unrotated fit on the intercept or time slope in any design. It did not (Table S1); the lowest design was 5.5 times and the lowest single seed 3.2 times. Per gradient evaluation the gains were 16.7 for the intercept and 22.0 for the time slope (geometric means over all designs), and the rotated fit took 0.81 times as many leapfrog steps per iteration.

The design with the fewest subjects and the most information per subject gained least,

Table S1: Stress grid: ESS-per-second ratios, rotated over unrotated (geometric means over 3 seeds).

visits	N	ρ	intercept	time slope	α
35	50	0.30	5.5	10.3	1.60
35	50	0.95	21.1	21.6	2.42
35	100	0.30	14.5	27.1	1.58
35	100	0.95	26.0	28.6	1.73
9	100	0.95	7.0	7.5	0.74
17	300	0.95	24.6	29.7	1.09

as the fixed-metric residual predicts. Strong correlation did not reduce the rotation’s advantage: at $\rho = 0.95$ the gains were as large as or larger than at $\rho = 0.3$, partly because the unrotated fit degrades faster. Nor did it hurt the rotated fit in absolute terms: with 35 visits its ESS per draw for the intercept was 1.25 at $\rho = 0.95$ against 0.68 at $\rho = 0.3$ with $N = 50$, and 1.50 against 1.18 with $N = 100$. The predicted weakness is visible only against a variant that removes the absorbable directions from the likelihood’s coordinates and restores the reported coefficients with an exact correction (Section 9.1). That variant has no fixed-metric residual, and in the 35-visit designs it was 1.4–2 times more efficient than the rotation, while being about equal to it with 9 visits. Posterior means agreed with the unrotated fit’s to a median of 0.029 posterior standard deviations (maximum 0.166, 90 pairs), there were no divergences, and wall time was 0.86–1.08 times the unrotated fit’s (geometric means per design). α , which the degeneracy does not involve, fell to 0.74 in the 9-visit design.

S4 Calibration

With $N = 300$ and 100 replicates, every one of which passed an $\hat{R} \leq 1.05$ check on the regression parameters (so none was excluded), coverage of 95% intervals agreed between the unrotated and rotated fits within 0.01, and the ratio of mean posterior standard deviation to the standard deviation of posterior means within 0.015, for every estimand (Table S2); the Monte Carlo standard error of a 95% coverage is about 0.011. Two features appear in both

arms and belong to the model and design: slightly wide intervals for β_0 , and an upward bias of about 3 Monte Carlo standard errors (2.8%) in α . The latter is consistent with ordinary finite-sample bias. In a separate study (`dev/study_alpha_bias.R`, Section S6), neither the spline baseline hazard nor the simulator’s hazard calibration explained it (paired changes of -0.0011 and -0.0016 , standard error about 0.0025), and at $N = 1200$ it fell to 1.1%, 1.7 Monte Carlo standard errors, with coverage 0.92.

Table S2: Calibration, $N = 300$, 100 replicates: coverage of 95% intervals, ratio of mean posterior SD to the SD of posterior means, and bias in Monte Carlo standard errors.

estimand	unrotated			rotated		
	cover	SD ratio	bias/se	cover	SD ratio	bias/se
β_0	0.99	1.164	1.68	0.99	1.150	1.55
β_{time}	0.97	1.027	-0.30	0.96	1.038	-0.32
α	0.93	0.922	2.97	0.93	0.917	2.97
σ_0	0.94	0.914	0.63	0.95	0.906	0.68

S5 Real data with a random intercept only

Fitted with a random intercept only, the two datasets of Section 8 show the degeneracy more starkly. Unrotated, β_0 and the subject-constant coefficients mixed at 0.04–0.07 ESS per draw, and 91–95% of their posterior variance was explained by the absorbable directions of the random effects; every other parameter mixed near one ESS per draw. The rotation raised the affected coefficients’ ESS per second 51–53 times on `aids` and 19–20 times on `pbc2` (2 seeds each), and the slowest parameter’s ESS per draw from 0.05 to 0.88 and from 0.06 to 0.78.

S6 Reproducibility

All code is in the code archive, which contains the `jmjax` package source. The development scripts below reproduce each result; paths are relative to the package root.

result	script (output)
Figure 1, Section 4.4 numbers	dev/theory_rotation_toy.py; dev/figures/make_figures.py
Section S1	dev/theory_alpha_toy.py
Section 5, R^2 and its split	dev/alpha_missing_information.R
Figure 2	dev/figures/make_figures.py
Table 1	dev/pilot_rotate_grid.R (dev/pilot_rotate_grid.csv)
Section 6.3 and Section S2, standard length	ROT_GRID=q1 dev/pilot_rotate_grid.R (dev/pilot_rotate_q1.csv)
Section S2	dev/study_q1_longrun.R (dev/study_q1_longrun.csv)
Table S1	ROT_GRID=stress dev/pilot_rotate_grid.R (dev/pilot_rotate_stress.csv)
Figure 3	dev/figures/make_figures.py
Table S2	dev/study_calibration.R
α bias, $N = 1200$	dev/study_alpha_bias.R (dev/alpha_bias_results.csv)
Table 2	dev/generalality/mixed_models.py (dev/generalality/results_orthodont_toenail_ext.csv)
Table 3	dev/study_realdata_rotate.R
Table 4; JMbayer2 in Figure 2	dev/study_common_ess.R (dev/common_ess.csv)
Section S5	dev/pilot_q1_intercept.R
α against JMbayer2, 1–12 covariates	dev/study_orth_vs_jmbayer2.R
Example 1, basis checks	dev/verify_absorbable_basis.py